THE DENOMINATORS OF A REGULAR MEROMORPHIC SECTION FORM A QUASI-COHERENT IDEAL WHOSE QUOTIENTS OF THE STRUCTURE SHEAF AND THE LINE BUNDLE ARE NOWHERE DENSE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Statement span 6172-6194 and proof span 6196-6217 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof contains the author's italicised emphasis on the word 'both' in the reduction step; it is preserved.

Allowed standard prerequisites (editorial).

- The sheaf of meromorphic sections of an invertible module and the fact that the sections of the module are contained in the meromorphic ones, so that the defining formula makes sense. Used at statement lines 6181-6186 and in the identification of the local situation at proof lines 6197-6199. Justifies the definition of the ideal of denominators and lets the proof pass to an affine chart where the bundle is trivial.
- A regular meromorphic section becomes, after shrinking, a quotient of two nonzerodivisors in the coordinate ring. Used at proof lines 6199-6200. Reduces the whole statement to an elementary statement about a fraction a over b in a ring with a and b nonzerodivisors.
- Quasi-coherence of a sheaf of ideals is checked on localizations, and the closed subset defined by a nonzerodivisor is nowhere dense in an integral or reduced setting. Used at proof lines 6202-6210 and 6211-6215. Gives quasi-coherence from the localization computation and the nowhere dense support of the two cokernels.

Lemma 0.1. *Let X be a scheme. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Let s be a regular meromorphic section of $\mathcal{L}$. Let us denote $\mathcal{I} \subset \mathcal{O}_X$ the sheaf of ideals defined by the rule*

$$\mathcal{I}(V) = \{f \in \mathcal{O}_X(V) \mid fs \in \mathcal{L}(V)\}.$$

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The formula makes sense since $\mathcal{L}(V) \subset \mathcal{K}_X(\mathcal{L})(V)$. *Then* $\mathcal{I}$ *is a quasi-coherent sheaf of ideals and we have injective maps*

$$1 : \mathcal{I} \longrightarrow \mathcal{O}_X, \quad s : \mathcal{I} \longrightarrow \mathcal{L}$$

whose cokernels are supported on closed nowhere dense subsets of X.

Proof. The question is local on X. Hence we may assume that $X = \mathrm{Spec}(A)$, and $\mathcal{L} = \mathcal{O}_X$. After shrinking further we may assume that $s = a/b$ with $a, b \in A$ *both* nonzerodivisors in A. Set $I = \{x \in A \mid x(a/b) \in A\}$.

To show that $\mathcal{I}$ is quasi-coherent we have to show that $I_f = \{x \in A_f \mid x(a/b) \in A_f\}$ for every $f \in A$. If $c/f^n \in A_f$, $(c/f^n)(a/b) \in A_f$, then we see that $f^m c(a/b) \in A$ for some m, hence $c/f^n \in I_f$. Conversely it is easy to see that I_f is contained in $\{x \in A_f \mid x(a/b) \in A_f\}$. This proves quasi-coherence.

Let us prove the final statement. It is clear that $(b) \subset I$. Hence $V(I) \subset V(b)$ is a nowhere dense subset as b is a nonzerodivisor. Thus the cokernel of 1 is supported in a nowhere dense closed set. The same argument works for the cokernel of s since $s(b) = (a) \subset sI \subset A$. □